

YEAR 12

TERM 2

MATHEMATICS EXAMINATION

MAY 2010

Total Time: 70 minutes
Total Marks: 60 marks

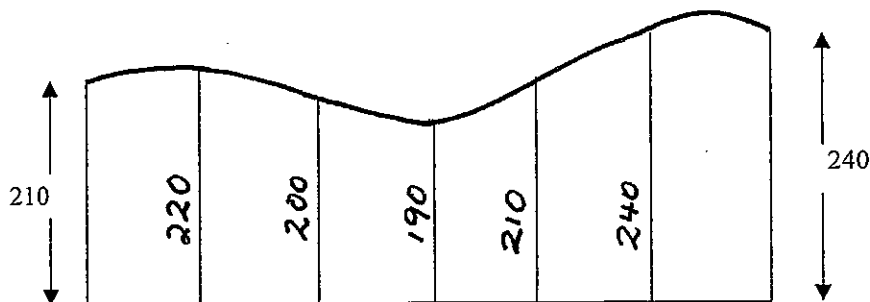
Directions to Candidates:

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board approved calculators may be used.
- Answer each question in a SEPARATE writing booklet.
- You may ask for extra writing booklets if you need them.

SECTION I – Integration (24 marks)

Marks:

- (a) (i) Find $\int 5. dx$ 1
- (ii) Find $\int \frac{3}{(x-6)^2} dx$ 2
- (iii) Find $\int \frac{5x^3 - 8}{x^2} dx$ 2
- (iv) Evaluate $\int_1^4 x^2 + \sqrt{x}. dx$ 3
- (b) (i) On a number plane, sketch the curve $y = 4x - x^2$ showing its important features. 2
- (ii) On the same diagram shade the region R, which is bounded by the curve $y = 4x - x^2$ and the line $y = 5$, between $x = 0$ and $x = 4$ 1
- (iii) Find the area of the shaded region R. 4
- (c) The diagram shows a block of land and its dimensions, in metres. The block of land is bounded, on one side, by a river. Measurements are taken perpendicular to the line AB, from AB to the river, at equal intervals of 50m.



4

Use Simpson's Rule with six subintervals to find an approximation to the area of the block of land.

- (d) (i) The area under the curve $y = \sqrt{9 - x^2}$, $-3 \leq x \leq 3$, is rotated about the X axis. Find the volume of the solid of revolution leaving answer in terms of π . 4
- (ii) Describe the shape of this solid. 1

Section II over page

SECTION III – Exponentials and Logarithmic Functions (36 marks)**Marks:**

(a) (i) Evaluate $\log_2 10$, correct to 3 decimal places. 2

(ii) Solve for x : $\log_9 x = \frac{-3}{2}$, giving your answer in exact form. 2

(b) Differentiate each of the following:

(i) $4e^{-2x}$ 1

(ii) $x^3 e^{3x}$ 2

(iii) $\ln(3x^2 - 7x + 2)$ 2

(iv) $[\ln(2x - 1)]^4$ 2

(v) $\ln\left(\frac{3-4x}{2x+3}\right)$ 2

(c) Find the following integrals :

(i) $\int \frac{8dx}{2x-1}$ 1

(ii) $\int 1 + \frac{3}{x-2} dx$ 2

(iii) $\int e^{5-2x} dx$ 1

(iv) $\int \frac{4x+2}{x^2+x} dx$ 2

(d) (i) Show that $\frac{d}{dx} e^{2x^4} = 8x^3 e^{2x^4}$ 1

(ii) Hence, or otherwise, find $\int x^3 \cdot e^{2x^4} dx$ 1

(e) Find the value of 'k', such that $\int_1^k \frac{4}{x^2} dx = 3$ 3

(f) Find the equation of the tangent to the curve $y = e^{\frac{x}{2}}$ at the point where
where $(1, e^{\frac{1}{2}})$ 3

(g) Continued over page

Marks:

(g) For the curve $y = x \log_e x$, $x > 0$,

- | | | |
|-------|---|---|
| (i) | find its X intercept(s) | 1 |
| (ii) | find any stationary point(s), and determine their nature | 3 |
| (iii) | determine if there are any points of inflexion on the curve | 1 |
| (iv) | find y when $x = e$ | 1 |
| (vi) | draw a neat sketch of the curve $y = x \log_e x$ showing its important features | 3 |

END OF ASSESSMENT

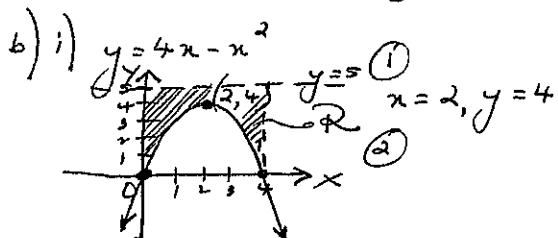
SECTION I - Integration (24 marks)SOLUTIONS
+ SUGGESTED
MARKING
SCALE

a) i) $\int 5 \cdot dx = 5x + C$

ii) $\int \frac{3}{(x-6)^2} dx = \int 3(x-6)^{-2} dx$
 $= 3 \frac{(x-6)^{-1}}{-1} + C$
 $= -\frac{3}{x-6} + C$ (2)

iii) $\int \frac{5x^3 - 8}{x^2} dx = \int \frac{5x^3}{x^2} - \frac{8}{x^2} dx$
 $= \int 5x - 8x^{-2} dx$
 $= \frac{5x^2}{2} - \frac{8x^{-1}}{-1} + C$
 $= \frac{5x^2}{2} + \frac{8}{x} + C$ (2)

iv) $\int_1^4 x^2 + \sqrt{x} \cdot dx = \int_1^4 x^2 + x^{1/2} \cdot dx$
 $= \left[\frac{x^3}{3} + \frac{2x^{3/2}}{3} \right]_1^4$
 $= \frac{4^3}{3} + \frac{2 \times 4^{3/2}}{3} - \left(\frac{1^3}{3} + \frac{2 \times 1^{3/2}}{3} \right)$
 $= \frac{64}{3} + \frac{16}{3} - \left(\frac{1}{3} + \frac{2}{3} \right)$
 $= \frac{77}{3}$ (3)



iii) R = rectangle - area under curve
 $= 5 \times 4 - \int_0^4 4x - x^2 dx$ (4)
 $= 20 - \left[2x^2 - \frac{x^3}{3} \right]_0^4$
 $= 20 - \left[2 \times 4^2 - \frac{4^3}{3} - 0 \right]$
 $= 20 - \left[32 - \frac{64}{3} \right] = 20 - 10\frac{2}{3} = 9\frac{2}{3}$

Section 1 cont...

c) $A = \frac{1}{3} [y_1 + 4y_2 + 2y_3 + \dots + y_n]$
 $= \frac{50}{3} [210 + 4 \times 220 + 2 \times 200 + 4 \times 190$
 $\quad + 2 \times 210 + 4 \times 240 + 240]$
 $= 64500 \text{ m}^2$ (4)

d) i) $y = \sqrt{9-x^2} \quad -3 \leq x \leq 3$

$$V = \pi \int_{-3}^3 y^2 \cdot dx$$
$$= \pi \int_{-3}^3 9-x^2 \cdot dx$$
$$= 2\pi \int_0^3 9-x^2 \cdot dx$$
$$= 2\pi \left[9x - \frac{x^3}{3} \right]_0^3$$
$$= 2\pi \left[9 \times 3 - \frac{3^3}{3} - 0 \right]$$
$$= 2\pi [27-9]$$
$$= 36\pi \text{ units}^3$$
 (4)

ii) a sphere, radius 3,
centre = origin (1)

SECTION II 36

$$a) i) \log_2 10 = \frac{\log_e 10}{\log_e 2} = 3.32192.. \quad (2)$$

$$\log_e 2 \div 3.322 \text{ (3 d.p.)}$$

$$ii) \log_9 x = -\frac{3}{2}$$

$$x = 9^{-3/2} = \frac{1}{(9)^3} = \frac{1}{27} \div 0.037 \quad (2)$$

$$b) i) \frac{d}{dx} 4e^{-2x} = -2 \cdot 4e^{-2x} = -8e^{-2x} \quad (1)$$

$$ii) \frac{d}{dx} x^3 e^{3x} = VU' + UV'$$

$$= e^{3x} \cdot 3x^2 + x^3 \cdot 3e^{3x}$$

$$= 3x^2 e^{3x} (1 + x) \quad (2)$$

$$iii) \frac{d}{dx} \ln(3x^2 - 7x + 2) = \frac{6x - 7}{3x^2 - 7x + 2} \quad (2)$$

$$iv) \frac{d}{dx} [\ln(2x-1)]^4 = 4 [\ln(2x-1)]^3 \cdot \frac{2}{2x-1}$$

$$= \frac{8 [\ln(2x-1)]^3}{2x-1} \quad (2)$$

$$v) \frac{d}{dx} \ln \left(\frac{3-4x}{2x+3} \right) = \frac{d}{dx} \ln(3-4x) - \ln(2x+3)$$

$$= \frac{-4}{3-4x} - \frac{2}{2x+3}$$

$$= \frac{-8x-12-6+8x}{(3-4x)(2x+3)} = \frac{-18}{(3-4x)(2x+3)} \quad (2)$$

$$c) i) \int \frac{8dx}{2x-1} = 4 \int \frac{2dx}{2x-1}$$

$$= 4 \ln(2x-1) + C \quad (1)$$

$$ii) \int 1 + \frac{3}{x-2} dx = x + 3 \ln(x-2) + C \quad (2)$$

$$iii) \int e^{5-2x} dx = \frac{e^{5-2x}}{-2} + C \quad (1)$$

SECTION II cont...

$$iv) \int \frac{4x+2}{x^2+x} dx = 2 \int \frac{2x+1}{x^2+x} dx$$

$$= 2 \ln(x^2+x) + C \quad (2)$$

$$d) i) \frac{d}{dx} e^{2x^4} = e^{2x^4} \cdot \frac{d}{dx} (2x^4)$$

$$= e^{2x^4} \cdot 8x^3$$

$$= 8x^3 \cdot e^{2x^4} \quad (1)$$

$$iii) \int x^3 \cdot e^{2x^4} dx = \frac{1}{8} e^{2x^4} + C \quad (1)$$

$$e) \int_1^k \frac{4}{x^2} = 3$$

$$LHS = \int_1^k 4x^{-2} dx$$

$$= \left[\frac{4x^{-1}}{-1} \right]_1^k$$

$$= \left[-\frac{4}{x} \right]_1^k$$

$$\therefore -\frac{4}{k} - \left(-\frac{4}{1} \right) = 3$$

$$-\frac{4}{k} + 4 = 3$$

$$1 = \frac{4}{k}$$

$$\therefore k = 4. \quad (3)$$

$$f) y = e^{x/2} \quad (1, e^{1/2}) \therefore \text{tangent is}$$

$$y - y_1 = m(x - x_1)$$

$$y - \sqrt{e} = \frac{\sqrt{e}}{2} (x - 1)$$

$$2y - 2\sqrt{e} = \sqrt{e}x - \sqrt{e}$$

$$\sqrt{e}x - 2y + \sqrt{e} = 0. \quad (3)$$

Q) $y = x \log_e x, x > 0$

i) x intercept $\Rightarrow y = 0$

$$0 = x \log_e x$$

$$x \neq 0 \therefore \log_e x = 0$$

$$\therefore x = 1$$

(1)

ii) st. pts when $y' = 0$

$$y' = x \cdot \frac{1}{x} + \log_e x \cdot 1$$

$$= 1 + \ln x$$

$$1 + \ln x = 0$$

$$\ln x = -1$$

$$\text{or } \log_e x = -1$$

$$\therefore x = e^{-1} = \frac{1}{e}$$

$$x = \frac{1}{e}, y = \frac{1}{e} \log_e \frac{1}{e}$$

$$= \frac{1}{e} \times -1$$

$$= -\frac{1}{e}$$

$$\therefore \left(\frac{1}{e}, -\frac{1}{e}\right)$$

test for nature y'' ,

$$y'' = \frac{1}{x}$$

$$f''\left(\frac{1}{e}\right) > 0 \therefore \text{min pt at } \left(\frac{1}{e}, -\frac{1}{e}\right)$$

(3)

iii) pts of inflex, $y'' = 0$

$$\frac{1}{x} = 0 \therefore \text{no solution}$$

$$1 = 0 ?? \rightarrow$$

$\therefore$ no pts. of inflexion. (1)

iv) $x = e, y = e \log_e e = e \times 1 = e$

$$\therefore (e, e)$$

(1)

